

# Heitor Ernandes, Ph.D.



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<https://github.com/drheitor>

## Education

- 2018 – 15/09/2022 **Ph.D., Universidade de São Paulo** Astronomy.  
Thesis title: *Abundances in field and globular cluster stars of the Galactic bulge.* (Appointed by the institute as the best thesis in Astronomy in 2022)  
Advisor: Beatriz Barbuy  
Included 1-year research secondment at the University of Edinburgh  
Advisor: Chris Evans
- 2013 – 2018 **B.Sc. Universidade de São Paulo** in Physics.

## Career History

- 05/12/2025 – **Postdoc., CAMK-PAN** Astronomy.  
Research Group led by Rodolfo Smiljanic  
Fellowship from the DAINA project between Poland and Lithuania
- 01/11/2022 – 01/12/2025 **Postdoc., Lunds Universitet** Astronomy.  
Research Group led by Sofia Feltzing  
Fellowship from AKW - SELTIC

## Visiting Scholar

- 2026- **Lund University** - Visiting research fellow - Sweden
- 2024-1 week **University of Ljubljana** - Host: Gregor Travern.  
Funding: Fysiografen - Sweden
- INAF-Osservatorio Astronomico di Trieste** - Host: Gabriele Cescutti.  
Funding: Fysiografen - Sweden
- 2023-1 week **MIAPbP** - Stellar Astrophysics in the era of Gaia, Spectroscopic, and Asteroseismic surveys
- 2023-2 weeks **MIAP-Heidelberg** - Host: Maria Bergemann.  
Funding: MIAP
- 2018-2 weeks **University of Edinburgh/STFC**, Host: Chris Evans.  
Funding: UK-Brazil Global Challenges Research Fund
- 2017-1 month **ESO-Chile** - Host: Bruno Dias.  
Funding: ESO

## Participation in the scientific teams

- 2025- **Participation in the HRMOS**  
Nucleo-chronology working group

## Participation in the scientific teams (continued)

- **Permanent member of 4MOST**
- 2022- ■ **Member of the Swedish ELT Instrumentation Consortium (SELTIC)**
- **Participation in the scientific team of 4MOST**  
As part of the 4MIDABLE (S4) Survey and High resolution Halo (S2) Survey
- **Participation in the 4MOST working groups**  
As part of the IWG<sub>1</sub> - Targeting Support and PCWG - Project Culture Working Group
- **Participation in the scientific team of the ANDES spectrograph**  
Developing the Data Analysis Software in Uppsala
- 2019- ■ **Participation in the scientific team of the CUBES spectrograph**  
Building the stellar astrophysics science case
- **Participation in the scientific team of the MOSAIC spectrograph**  
Working on the COMPASS simulations for metal-poor stars

## Research Publications

264 citations - h-index = 11 [ADS link](#).

330 citations - h-index = 12 [Google Scholar link](#).

### Journal Articles

- 1 Ernandes, H., Barbuy, B., Chiappini, C., Feltzing, S., Pérez-Villegas, A., Friaça, A. C. S., ... Plotnikova, A. (2026). Abundances in 78 metal-rich bulge spheroid stars from APOGEE., 707, A328.  
[doi:10.1051/0004-6361/202558789](#). arXiv: 2602.12415 [astro-ph.GA]
- 2 Ortolani, S., Souza, S. O., Nardiello, D., Barbuy, B., Bica, E., Ferreira, B. P. L., ... Ernandes, H. (2026). Ancient Relic Moderately Metal-rich Bulge Cluster Tonantzintla 2., 998(1), L8.  
[doi:10.3847/2041-8213/ae3b3e](#). arXiv: 2601.18567 [astro-ph.GA]
- 3 Ernandes, H., Skúladóttir, Á., Feltzing, S., & Feuillet, D. (2025). Disentangling Milky Way halo populations at low metallicities using [Al/Fe]., 703, A256. [doi:10.1051/0004-6361/202557149](#). arXiv: 2509.06773 [astro-ph.GA]
- 4 Kordopatis, G., Feuillet, D., Lehmann, C., Feltzing, S., Minchev, I., Hill, V., & Ernandes, H. (2025). First large-scale spatial and velocity patterns of local metal-rich stars in the Milky Way., 703, A151.  
[doi:10.1051/0004-6361/202556657](#). arXiv: 2509.17420 [astro-ph.GA]
- 5 Barbuy, B., Fernández-Trincado, J. G., Camargo, M. S., Geisler, D., Brauner, M., Villanova, S., ... Pignatari, M. (2025). Investigating Phosphorus Abundances in a Sample of APOGEE-2 Bulge Globular Clusters., 170(4), 245. [doi:10.3847/1538-3881/ae046b](#)
- 6 Barbuy, B., Ernandes, H., Friaça, A. C. S., Camargo, M. S., da Silva, P., Souza, S. O., ... Nitschelm, C. (2025). Abundances of P, S, and K in 58 bulge spheroid stars from APOGEE., 700, A184.  
[doi:10.1051/0004-6361/202555713](#). arXiv: 2507.11667 [astro-ph.SR]
- 7 Skúladóttir, Á., Ernandes, H., Feuillet, D. K., Mori, A., Feltzing, S., Lucchesi, R. E. R., & Di Matteo, P. (2025). Evidence of Gaia Enceladus Experiencing at Least Two Passages around the Milky Way., 986(2), L21. [doi:10.3847/2041-8213/addc66](#). arXiv: 2506.00409 [astro-ph.GA]
- 8 Barbuy, B., Friaça, A. C. S., & Ernandes, H. (2025). Nucleosynthesis of cobalt and copper. *Galaxies*, 13(5).  
[doi:10.3390/galaxies13050113](#)

- 9 Barbuy, B., Friaça, A. C. S., Ernandes, H., da Silva, P., Souza, S. O., Fernández-Trincado, J. G., ... Nitschelm, C. (2024). Abundances of iron-peak elements in 58 bulge spheroid stars from APOGEE., *691*, A296. [doi:10.1051/0004-6361/202452235](#). arXiv: 2410.13751 [astro-ph. SR]
- 10 **Ernandes, H.**, Feuillet, D., Feltzing, S., & Skúladóttir, Á. (2024). Gaia-Sausage-Enceladus star formation history as revealed by detailed elemental abundances: An archival study using SAGA data., *691*, A333. [doi:10.1051/0004-6361/202450827](#). arXiv: 2405.13641 [astro-ph. GA]
- 11 da Silva, P., Barbuy, B., Ernandes, H., Souza, S. O., Fernández-Trincado, J. G., & González-Díaz, D. (2024). Abundances in eight bulge stars from the optical and near-infrared. *arXiv e-prints*, arXiv:2405.06153. [doi:10.48550/arXiv.2405.06153](#). arXiv: 2405.06153 [astro-ph. SR]
- 12 Barbuy, B., Friaça, A. C. S., Ernandes, H., Moura, T., Masseron, T., Cunha, K., ... Razera, R. (2023). Light elements Na and Al in 58 bulge spheroid stars from APOGEE., *526(2)*, 2365–2376. [doi:10.1093/mnras/stad2888](#)
- 13 **Ernandes, H.**, Castro, M. J., Barbuy, B., Spite, M., Hill, V., Castilho, B., & Evans, C. J. (2023). Reanalysis of neutron-capture elements in the benchmark r-rich star CS 31082-001. *arXiv e-prints*, arXiv:2306.17627. arXiv: 2306.17627 [astro-ph. SR]
- 14 Souza, S. O., **Ernandes, Heitor**, Valentini, M., Barbuy, B., Chiappini, C., Pérez-Villegas, A., ... Bica, E. (2023). Chrono-chemodynamical analysis of the globular cluster NGC 6355: Looking for the fundamental bricks of the Bulge. *arXiv e-prints*, arXiv:2301.05227. arXiv: 2301.05227 [astro-ph. GA]
- 15 Evans, C., Cristiani, S., Opitom, C., Cescutti, G., D’Odorico, V., Alcalá, J. M., ... Zafar, T. (2022). The CUBES science case. *Experimental Astronomy*. [doi:10.1007/s10686-022-09864-7](#). arXiv: 2208.01677 [astro-ph. IM]
- 16 Razera, R., Barbuy, B., Moura, T. C., **Ernandes, H.**, Pérez-Villegas, A., Souza, S. O., ... Zoccali, M. (2022). Abundance analysis of APOGEE spectra for 58 metal-poor stars from the bulge spheroid. [doi:10.1093/mnras/stac2136](#). arXiv: 2208.06634 [astro-ph. SR]
- 17 Zanutta, A., Cristiani, S., Atkinson, D., Baldini, V., Balestra, A., Barbuy, B., ... Zorba, S. (2022). CUBES phase a design overview. *Experimental Astronomy*. [doi:10.1007/s10686-022-09837-w](#). arXiv: 2203.15352 [astro-ph. IM]
- 18 **Ernandes, H.**, Barbuy, B., Friaça, A., Hill, V., Spite, M., Spite, F., ... Evans, C. J. (2022). Be, V, and Cu in the halo star CS 31082-001 from near-UV spectroscopy., *510(4)*, 5362–5375. [doi:10.1093/mnras/stab3789](#). arXiv: 2202.04450 [astro-ph. SR]
- 19 **Ernandes, H.**, Barbuy, B., Castilho, B., Evans, C. J., & Cescutti, G. (2022). Simulated observations of heavy elements with CUBES. *Experimental Astronomy*. [doi:10.1007/s10686-021-09829-2](#). arXiv: 2203.15693 [astro-ph. IM]
- 20 Barbuy, B., **Ernandes, H.**, Souza, S. O., Razera, R., Moura, T., Meléndez, J., ... Bica, E. (2021). Gemini/Phoenix H-band analysis of the globular cluster AL 3. *A&A*, *648*, A16. [doi:10.1051/0004-6361/202039761](#). arXiv: 2102.12674 [astro-ph. SR]
- 21 **Ernandes, H.**, Barbuy, B., Friaça, A. C. S., Hill, V., Zoccali, M., Minniti, D., ... Ortolani, S. (2020). Cobalt and copper abundances in 56 Galactic bulge red giants. *A&A*, *640*, A89. [doi:10.1051/0004-6361/202037869](#). arXiv: 2007.00397 [astro-ph. SR]
- 22 **Ernandes, H.**, Dias, B., Barbuy, B., Kamann, S., Ortolani, S., Cantelli, E., ... Rossi, L. (2019). A MUSE study of the inner bulge globular cluster Terzan 9: a fossil record in the Galaxy. *A&A*, *632*, A103. [doi:10.1051/0004-6361/201936431](#). arXiv: 1910.09893 [astro-ph. GA]
- 23 Barbuy, B., Muniz, L., Ortolani, S., **Ernandes, H.**, Dias, B., Saviane, I., ... Held, E. V. (2018). High-resolution abundance analysis of four red giants in the globular cluster NGC 6558. *A&A*, *619*, A178. [doi:10.1051/0004-6361/201833953](#). arXiv: 1810.11703 [astro-ph. SR]

- 24 **Ernandes, H.**, Barbuy, B., Alves-Brito, A., Friaça, A., Siqueira-Mello, C., & Allen, D. M. (2018). Iron-peak elements Sc, V, Mn, Cu, and Zn in Galactic bulge globular clusters. *A&A*, 616, A18. [doi:10.1051/0004-6361/201731708](https://doi.org/10.1051/0004-6361/201731708). arXiv: 1801.06157 [astro-ph.SR]
- 25 Barbuy, B., Cantelli, E., Vemado, A., **Ernandes, H.**, Ortolani, S., Saviane, I., ... Siqueira-Mello, C. (2016). High-resolution abundance analysis of red giants in the metal-poor bulge globular cluster HP 1. *A&A*, 591, A53. [doi:10.1051/0004-6361/201628106](https://doi.org/10.1051/0004-6361/201628106). arXiv: 1604.02095 [astro-ph.SR]

## Conference Proceedings

- 1 Cristiani, S., Alcalá, J. M., Alencar, S. H. P., Balashev, S., Bastian, N., Barbuy, B., ... Zanutta, A. (2022). CUBES: the Cassegrain U-band Efficient Spectrograph. In C. J. Evans, J. J. Bryant, & K. Motohara (Eds.), *Ground-based and airborne instrumentation for astronomy ix* (Vol. 12184, 12184oA). [doi:10.1117/12.2629990](https://doi.org/10.1117/12.2629990). arXiv: 2208.01672 [astro-ph.IM]
- 2 **Ernandes, H.**, Evans, C. J., Barbuy, B., Castilho, B., Cescutti, G., Christlieb, N., ... Smiljanic, R. (2020). Stellar astrophysics in the near-UV with VLT-CUBES. In *Society of photo-optical instrumentation engineers (SPIE) conference series* (Vol. 11447, p. 114476o). [doi:10.1117/12.2562497](https://doi.org/10.1117/12.2562497). arXiv: 2102.02205 [astro-ph.IM]
- 3 **Ernandes, H.**, Dias, B., Barbuy, B., Kamann, S., Ortolani, S., Rossi, L., ... Bica, E. (2018). A MUSE study of the inner bulge globular cluster Terzan 9 : a fossil record in the Galaxy. In *The galactic bulge at the crossroads* (p. 62). [doi:10.5281/zenodo.2651064](https://doi.org/10.5281/zenodo.2651064)

## Invited Seminars

- 2026  **Astronomical Observatory of the University of Warsaw -"GSE star formation history as revealed by detailed elemental abundances "**
- 2024  **Seminar in University of Ljubljana -"GSE star formation history as revealed by detailed elemental abundances "**
-  **Seminar in INAF-Osservatorio Astronomico di Trieste -"GSE star formation history as revealed by detailed elemental abundances "**
- 2023  **Seminar in Stockholm University-"The Bright Future of the Ground-Based Near-UV Spectroscopy for r-rich stars"**

## Fellowships and Grants

- 2025  **Research Collaboration Grant** Fysiografen - The Fund of the Walter Gyllenberg Foundation  
Title: *Elemental Abundances Tracing the Gaia Sausage-Enceladus' Star Formation History*
-  **Research Collaboration Grant** KVA - Kungliga Vetenskapsakademien  
Title: *Elemental Abundances Tracing the Gaia Sausage-Enceladus' Star Formation History*
- 2024  **Travel Grant** Fysiografen - Young Researchers - Postdoctoral Fellows  
Title: *Neutron-capture Elements Tracing the Gaia-Sausage-Enceladus' Star Formation History*

## Fellowships and Grants (continued)

- 2023      **Research Collaboration Grant** Fysiografen - The Fund of the Walter Gyllenberg Foundation  
Title: *Unveiling Galactic Chemical Evolution with Non-LTE Analysis and Advanced Modelling*
- 2018 – 2022      **Ph.D., Universidade de São Paulo** CAPES.  
Thesis title: *Abundances in field and globular cluster stars of the Galactic bulge.*  
Advisor: Beatriz Barbuy
- 2019 – 2020      **Sandwich Ph.D., University of Edinburgh** PRINT-CAPES.  
Thesis title: *Abundances in field and globular cluster stars of the Galactic bulge.*  
Advisor: Chris Evans
- 2014 – 2017      **Undergraduate research project** PIBIC.  
*Abundância de elementos do pico do ferro em aglomerados globulares.*  
Advisor: Beatriz Barbuy

## Activities

- 2023 - 2026      **Outreach in The open nights at Lund Observatory**
- 2022 - 2026      **Group seminars** (Organizer-Chair)
-  **Referee papers**  
2026 1 MDPI, 1 Astrophysical Journal

## Congress, meeting, and workshops

- 2024      **Congress: IAU Symposium 395: Stellar Populations in the Milky Way and Beyond**, Paraty, Brazil (Talk)
-  **Workshop: Swedish e-Science Academy and Satellite Meetings 2024**, Lund, Sweden
- 2023      **Workshop: The Milky Way as Seen through Neutron-Capture Elements**, Lund, Sweden (Invited Speaker)
-  **Congress: MIAPbP on Stellar Astrophysics in the Era of Gaia, Spectroscopic, and Asteroseismic Surveys**, Munich, Germany
-  **Congress: EAS 2023**, Virtual (Poster)
-  **Congress: IAUS 379: Dynamical Masses of Local Group Galaxies**, Potsdam, Germany
- 2021      **Workshop: Workshop on the Future of MOS Technologies**, Virtual
- 2020      **School: JWST-UFRGS**, Virtual
-  **Meeting: Pristine Meeting**, Bern, Switzerland  
Presenting: CUBES and MOSAIC simulations (Talk)

## Congress, meeting, and workshops (continued)

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- 2018        **School: Dark Matter and Neutrinos**, São Paulo, Brazil
-     **Congress: IAU XXX**, Vienna, Austria
-     **Congress: The Galactic Bulge at the Crossroads**, Pucon, Chile  
         Presenting: Terzan 9: a MUSE analysis
- 2016        **Workshop: International Colloquium ASSOS**, São Paulo, Brazil
-     **Congress: Congress of Physics Paulo Leal**, São Paulo, Brazil
-     **School: School of Spectroscopy in Astrophysics and Laboratory Plasmas**, São Paulo, Brazil
- 2015        **Workshop: Workshop on Neutrino and Nuclear Astrophysics in IEA-USP**, São Paulo, Brazil

## Observation proposals and Honours

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- 2024        **coPI: Gemini Proposal. Cycle 32, ID. #GN2026A-FT-209**  
         MAROON-X Analysis of the metal-poor r-element-rich star TYC 3814-1598-1
- 2022        Appointed as the best PhD thesis in the astronomy division at IAG-USP
- 2024        **PI: HST Proposal. Cycle 32, ID. #17792**  
         STIS Analysis of the metal-poor r-element-rich star TYC 3814-1598-1